Interrupt vs Exception –

Interrupt:

CPU executing ⟶ External Event ⟶ Interrupt Signal ⟶ ISR Execution ⟶ Resume Program

-Triggered by Hardware external of CPU.
-**Asynchronous**: Independent of the CPU's current instruction execution.
-Examples: Timer Overflow, External GPIO Trigger, CAN Rx Interrupt, SPI Transfer Complete

Exception:

CPU executing → Fault/Condition Detected → Exception Trigger → Handler Execution → Resume/Terminate

-An event that occurs **during program execution** that disrupts the normal flow of instructions.
-**TRAP** = TriCore-specific implementation of Exception
-**Synchronous** : Tied to current CPU Instruction
-**Trap Classes (for Infineon TriCore Architecture):**

- Class 0: **MMU / Virtual Memory**
- Class 1: **Internal Protection**
- Class 2: **Instruction Errors**
- Class 3: **Context Management**
- Class 4: **Bus Errors**
- Class 5: **Arithmetic Errors**
- Class 6: **System Call**
- Class 7: **NMI / Reset**

Vector Parameter	Hardware Interrupts	Software / Processor Exceptions
Source	External to the processor silicon core (e.g., PCIe controllers, timers, GPIO pins, DMA controllers, PMICs).	Internal to the execution pipeline (e.g., ALU, MMU, Instruction Decoder).
Timing Relation	**Asynchronous:** Can strike at any arbitrary clock cycle. Independent of the current program counter (*PC*).	**Synchronous:** Bound exactly to the execution phase of the offending or invoking instruction. Reproducible.
Typical Root Causes	• Inbound network packets available • Hardware timer period expiration • Peripheral buffer full/empty status • External sensor threshold trigger	• Fatal conditions (Faults: Division by zero) • Memory management flags (Page Faults) • Intentional trap vectors (Syscalls) • Architectural violations (Illegal opcode)
Program Counter Position	Points to the *next* instruction that was scheduled to execute when the interrupt line asserted.	Points to either the exact instruction causing the fault, or the immediate following instruction (Trap).